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Accelerated intermittent theta-burst stimulation broadly ameliorates symptoms and cognition in Alzheimer's disease: A randomized controlled trial

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ABSTRACT

Background: Deficits in associative memory (AM) are the earliest and most prominent feature of Alzheimer's disease (AD) and demonstrate a clear cause of distress for patients and their families.
Objective: The present study aimed to determine AM enhancements following accelerated intermittent theta-burst stimulation (iTBS) in patients with AD.
Methods: In a randomized, double-blind, sham-controlled design, iTBS was administered to the left dorsolateral prefrontal cortex (DLPFC) of patients with AD for 14 days. Measurements included AM (primary outcome) and a comprehensive neuropsychological battery. Patients were evaluated at baseline, following the intervention (week 2), and 8 weeks after treatment cessation (week 10).
Results: Sixty patients with AD were initially enrolled; 47 completed the trial. The active group displayed greater AM improvements compared with the sham group at week 2 ($P = 0.003$), which was sustained at week 10. Furthermore, higher Mini-Mental State Examination (MMSE) scores at baseline were associated with greater AM improvements at weeks 2 and 10. For the independent iTBS group, this correlation predicted improvements in AM ($P < 0.001$) and identified treatment responders with 92% accuracy. Most of the neuropsychological tests were markedly improved in the active group. In particular, the Montreal Cognitive Assessment and MMSE in the active group increased by 2.8 and 2.3 points, respectively, at week 2, while there was no marked change in the sham group.
Conclusion: In the present study, accelerated iTBS of the DLPFC demonstrated an effective and well-tolerated complementary treatment for patients with AD, especially for individuals with relatively high MMSE scores.
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Abbreviations

AM	associative memory
AD	Alzheimer's disease
iTBS	intermittent theta-burst stimulation
DLPFC	dorsolateral prefrontal cortex
MMSE	Mini-Mental State Examination
rTMS	repetitive transcranial magnetic stimulation
CDR	Clinical Dementia Rating
MNI	Montreal Neurological Institute
ADL	Activities of Daily Living
HAMA	Hamilton Anxiety Rating Scale
HAMD	Hamilton Depression Rating Scale
NPI	Neuropsychiatric Inventory
LMT	Logical Memory Test
AVLT	Auditory Verbal Learning Test
DST-F/B	Digital Span Test (forward/backward)
SDMT	Symbol Digit Modalities Test
SCWT	Stroop Color-Word Test
CTT	Color Trail Test
CDT	Clock-Drawing Test
HVOT	Hooper Visual Organization Test
JLOT	Judgment of Line Orientation Test
BNT	Boston Naming Test
VFT-L/S	Verbal Fluency Test-letter and semantic

1. Introduction

Alzheimer's disease (AD) is a debilitating neurodegenerative disorder characterized by progressive cognitive impairment. Globally, it is a leading cause of disability in people over 65 years of age, for which there currently is no successful neuroprotective treatment [1]; therefore, AD poses a substantial socioeconomic concern [2]. Consequently, it is important to develop alternative therapeutic approaches to treat AD [3]. Repetitive transcranial magnetic stimulation (rTMS) approaches are relatively new to AD therapeutics; however, many studies have indicated that high-frequency (10–20 Hz) rTMS may be a potential alternative due to its positive cognitive effects on healthy individuals [4–6].

Loss of associative memory (AM), which is the binding of contextual information to stimulus attributes, is commonly considered an early and prominent feature of AD [7]. Measures of AM, which are most vulnerable to AD's pathological processes, are correlated with cerebrospinal fluid markers of AD neurodegeneration [8]. Thus, AM is a critical measure that reflects the efficacy of novel treatment of AD. In addition to the hippocampus and left inferior frontal gyrus, functional magnetic resonance imaging (MRI) studies have linked AM to the left dorsolateral prefrontal cortex (DLPFC) [9,10]. A vast number of studies suggest that regions of the DLPFC have a key role in AM proceedings, including encoding [11], binding [12], and retrieval [13]. The DLPFC is known to enhance binding through control operations [14] and strategy use [15,16], which facilitates item–item binding [17–20]. Leshikar et al. found that increasing the excitability of the DLPFC can improve AM by enhancing the encoding of details and strategy use that subsequently enable accurate recollection [21]. Moreover, older adults depend on the DLPFC even more than the hippocampus when intentionally learning associations [22,23]. Dysfunction of cortical plasticity of the DLPFC, which is induced by aging and the pathologic change of AD, may be a potential mechanism causing AM deficits in AD and aging [7,24,25]. Therefore, improving the excitability of the DLPFC and restoring its plasticity may be an effective way to enhance the AM of the elderly and in AD patients.

rTMS is an effective method to improve the excitability and plasticity of the cerebral cortex. rTMS can generate a time-varying magnetic field through an external coil and act on the cerebral cortex to induce current. The inducing orthogonal electrical current can cause neuroplastic changes that resemble either long-term potentiation (LTP) or depression. The enhanced LTP would benefit AM, especially its encoding [21]. Combining cognitive training, a recent study indicated that high frequency rTMS on the DLPFC could markedly improve the AM of AD patients. Whether rTMS is successful without cognition training is unknown. Based on the previous success of rTMS among healthy older adults [26], we hypothesize that high frequency rTMS on the DLPFC could also ameliorate AM deficits among patients with AD.

The sequence is the key factor affecting rTMS efficacy [27–31]. Intermittent theta-burst stimulation (iTBS) is a novel rTMS protocol that mimics endogenous theta rhythms, has more potent excitatory effects, and can improve the induction of synaptic LTP in a much shorter time than traditional stimulation (5 min versus 30 min) [30–33]. Prior studies show that compared with the traditional rTMS protocol, iTBS on the left DLPFC can improve cognitive processing more efficiently. Therefore, fewer cognitive resources were needed after iTBS to achieve a comparable behavioral performance [34–36]. A recent study identified that the use of "frequency-specific oscillations external stimulation," based on the brainwave frequency inherent to the brain area, significantly improved the function of the target [37,38]. Since the theta oscillations of the DLPFC have a key role in AM encoding [39,40], we hypothesize that using iTBS may improve AM function. Researchers have also suggested that individualized image navigation may lead to better aftereffects than the conventional scalp-landmark approach [41,42]. To date, there is a lack of randomized control trials (RCTs) investigating the efficacy of image-navigation iTBS to enhance AM and clinical symptoms among patients with AD.

The present study aimed to estimate the efficacy of iTBS to enhance AM among patients with AD using a randomized, double-blind, sham-controlled trial [43]. Comprehensive clinical and cognitive tests were conducted to identify the individualized outcome predictors, which were further validated by independent data.

2. Materials and methods

2.1. Standard protocol approvals, registrations, and patient consent

The current study protocol was reviewed and approved by the institutional ethics committee of Anhui Medical University. All participants provided written informed consent before participation. The first and last patients were enrolled on August 14, 2018 and January 3, 2020, respectively. Follow-up was completed by March 17, 2020. This study was registered with ClinicalTrials.gov (NCT03612622).

2.2. Design overview: sample size, randomization, and masking

This was a randomized, double-blind, parallel, placebo-controlled clinical trial assessing the efficacy of neuronavigational iTBS among patients with AD. The sample size was determined based on previous methodology [26], with 20 patients assigned to each arm, providing 80% power with an α level of 0.05.

Participants were divided into two groups using a random number generator. After baseline estimating (day 0), they received active or sham iTBS over the left DLPFC for 14 consecutive days (days 1–14). A researcher who was not involved in the assessments or stimulation performed the randomization of the patients. Treatment outcomes were estimated at day 15 (week 2) and 8 weeks after cessation of treatment (week 10).

The physicians and researcher performing the rTMS and randomization without patient contact, respectively, were aware of the group information. Both patients and clinical investigators remained blinded to the treatment procedure until the completion of the study. The patients were instructed not to discuss their treatment allocation with the staff or other patients to ensure a double-blind design.

2.3. Study participants

Patients with a clinical diagnosis of AD were recruited from the Memory Clinic of the First Affiliated Hospital of Anhui Medical University, Hefei, China.

Participants were included if they (1) met the core criteria for probable AD according to the National Institute of Neurological and Communicative Diseases and Stroke/Alzheimer's Disease and Related Disorders Association [44,45]; (2) scored 10–27 on the Mini-Mental State Examination (MMSE); (3) had a Clinical Dementia Rating (CDR) score $\leq$2; (4) were treated with a stable dose of donepezil for at least 3 months before iTBS treatment until the follow-up was completed [46]; and (5) were $\geq$50 years of age. Three senior-ranked clinicians (L.W., K.W., and Y.T.) determined the final diagnosis.

The exclusion criteria were as follows: (1) clinical features or work-up suggestive of an underlying pathology other than AD; (2) a history of significant head trauma or neurological disorders; (3) focal brain lesions on T1 or T2 images; (4) a history of seizures or history among first-degree relatives; (5) an implanted pacemaker, vagal stimulator, or deep brain stimulator; and (6) previous rTMS treatment.

2.4. Study interventions: MRI-navigated rTMS

The iTBS was performed using a Magstim Rapid^2 transcranial magnetic stimulator (Magstim, Whitland, UK) with a 70-mm air-cooled figure-of-eight coil. The iTBS parameters were as follows: 3 pulses, 50 Hz bursts given every 200 ms (at 5 Hz), and an intensity of 70% of the resting motor threshold (RMT) [33]. Because high-dose TBS may enhance aftereffects [47,48], we applied three rounds of iTBS on each treatment day (total: 1800 pulses/day $\times$ 14 days). The three iTBS sessions were separated by 15-min intervals [49]. To accurately target the coil placement, all stimulations were guided by the participant's anatomical image ($1 \times 1 \times 1$ mm^3) and a frameless neuro-navigation system (Brainsight; Rogue Research, Montreal, QC, Canada). The coil was placed with Visor 2.0 and was held tangentially to the skull with a handle pointing in the posterolateral direction. The junction point of the figure-of-eight Magstim coil was placed directly over the localized region.

The RMT was measured based on the stimulation intensity from the right first dorsal interosseous muscle using a handheld 70 mm figure-of-eight coil. RMT was defined as the lowest intensity required to evoked a small response (>50 μV) in more than five of ten consecutive trials in the right first dorsal interosseous muscle [36].

According to previous studies, the left DLPFC may be an effective target for patients with AD [26,50,51]. Therefore, we restricted our target within the left DLPFC, a sphere region centered on Montreal Neurological Institute (MNI) coordinates [−38 44 26] [52] with a 10-mm radius. We used SPM 12 (www.fl.ion.ucl.ac.uk/spm) and an in-house TMStarget Software [33] to transform the MNI coordinates into individual space.

Patients in the sham group were treated using a placebo coil (Magstim), which produced a similar sound and sensation on the scalp to the active coil but did not induce a sufficient current in the cortex. The patients were instructed not to discuss their treatment allocation with the staff or other patients to ensure a double-blind design.

2.5. Primary outcome

The AM was the only a-priori primary outcome, estimated using a face-cued word association test at baseline and repeated at the end of treatment (i.e., week 2). In the face-cued word association test, participants studied 12 human face photographs presented individually in grayscale on a computer screen for 4 s per face. Each photograph displayed a unique common word that the participants read aloud as each face-word pair was shown. The participants were instructed to memorize the word associated with each face. One minute after the face-word pairs were presented, participants were shown the same 12 faces, individually and in a different and randomized order. They were asked to recall the words that were presented with each face. Each face was scored as correct or incorrect, and the number of successful answers was defined as the AM score. Notably, the total duration of AM was relatively short (<5 min). Thus, the AM score may largely reflect the encoding function of participants, rather than storage or recall. There were four alternative versions of the test, and the order of each set was randomized across subjects using a Latin square. All participants performed the test twice with 3 min between runs, each in a different order. The mean number of correct responses between the two trials was considered the final score.

2.6. Secondary outcomes

The secondary outcomes included changes in clinical symptoms and multi-domain cognition tests at weeks 2 and 10. Adverse events (e.g., painful scalp sensations, twitching of the eyelid, tinnitus, epilepsy, or epileptic seizures) were recorded throughout the study by self-report.

Clinical symptom tests included the MMSE, Montreal Cognitive Assessment (MoCA, Beijing version), CDR, Lawton-Brody Activities of Daily Living (ADL) scale, Hamilton Anxiety Rating Scale with 14 items (HAMA), Hamilton Depression Rating Scale with 17 items (HAMD), Neuropsychiatric Inventory (NPI), and Chinese smell identification test.

The multi-domain cognition tests included the Logical Memory Test (LMT), Chinese version of the Auditory Verbal Learning Test (AVLT), Digital Span Test (forward/backward, DST-F/B), Symbol Digit Modalities Test (SDMT), Stroop Color-Word Test (SCWT), Color Trail Test (CTT) A/B, Clock-Drawing Test (CDT), Hooper Visual Organization Test (HVOT), Judgment of Line Orientation Test (JLOT), Boston Naming Test (BNT; both the BNT-spontaneous and BNT-total were adopted), and Verbal Fluency Test-letter and semantic (VFT-L/S).

2.7. Statistical analyses

2.7.1. Baseline characteristics and therapeutic efficacy

Baseline characteristics between the two groups were compared using the χ^2 test for categorical variables, and the t- or Mann-Whitney U tests for continuous variables.

Treatment outcomes were examined using linear mixed-effect models nested within individuals. Post-hoc analyses were

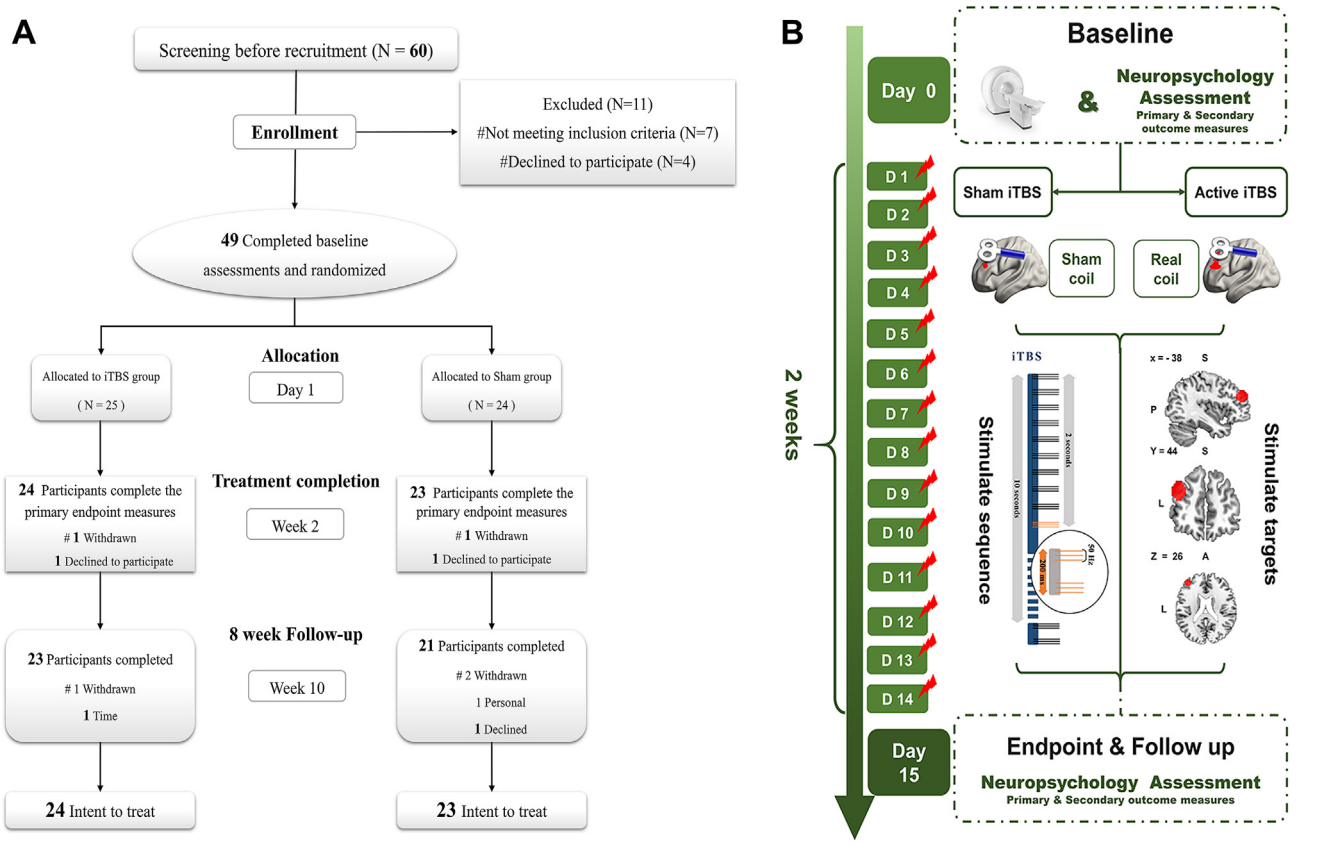


Fig. 1. The flowchart and diagrammatic representation of the timeline for the present study.

performed using Sidak's multiple comparison test. We defined a normalized change as the percentage (post-treatment versus baseline) relative to baseline. Treatment responders were participants with normalized AM changes larger than 50% at week 2.

2.7.2. Correlation and prediction analysis

Pearson's correlation analysis between AM and clinical variables was conducted to explore the independent predictor factor. We used a multivariate linear regression model with normalized AM changes at the end of treatment as the dependent factor, and baseline assessments as independent factors. Only measures significantly correlated with the AM at baseline were included in this model.

Logistic regression was used to identify responder predictors, where clinical characteristics were considered independent factors. The outcome of iTBS treatment (responder or non-responder) was the dependent factor.

2.7.3. Independent data validation for treatment prediction

To validate the prediction model, we included an independent cohort (n = 13) from our previous open-label iTBS study [43]. The stimulation protocol was the same as the active group in the current study. Briefly, the predicted AM changes of the independent data were computed using their baseline measures and the prediction model. Pearson's correlation was performed between the actual and predicted AM changes across participants.

All analyses were conducted using SPSS (version 26; IBM, Armonk, NY USA). All hypotheses were tested at a significance level of 0.05, using two-tailed tests.

3. Results

3.1. Characteristics of the study population

Of the 60 patients who completed screening, 49 were randomized to receive active (n = 25) or sham (n = 24) iTBS. Finally, 46 patients completed the follow-up (details in Fig. 1). When questioned on week 2, all participants believed that they received the active treatment.

3.2. Baseline measures

Baseline demographics, neuropsychological measures, and comorbidities (i.e., hypertension and diabetes mellitus) were well balanced between the active and sham groups (Table 1).

3.3. Primary outcome

There was an interaction effect between time (baseline and weeks 2 and 10) and group (active and sham) on AM scores ($F_{(2,87)}$ = 12.65, P < 0.001). AM scores improved markedly in the active group (from 1.56 to 2.54 with a 72.83% normalized change, t = 3.03, P = 0.012, effect size = 0.87; 95% CI, 0.25–1.49), but not the sham group (from 1.39 to 1.20 with a −10.46% normalized change, t = 1.12, P = 0.472) at week 2. Similar findings were observed at follow-up (week 10). Specifically, there was an improvement in AM among participants in the active group (from 1.56 to 2.28 with a 47.24% normalized change, t = 2.52, P = 0.038, effect size = 0.77; 95% CI, 0.12–1.34), but not the sham group (from 1.39 to 1.17 with a −4.99% normalized change, t = 0.65, P = 0.771, effect size = 0.20; 95% CI, −0.4–0.79; Table 2, Fig. 2A). Finally, the active group

Table 1
Baseline characteristics of patients who completed treatment.

Measures	Active	Sham	t/Z	p
Demographic characteristics				
Age (year)	66.46 (8.25)	66.35 (7.99)	0.05 [b]	0.963
Gender (male/female)	10/14	11/12	0.52[c]	0.564
Education years	9.71 (4.65)	9.52 (4.08)	−0.01[a]	0.991
APOE 4 carrier (%)	16/8 (66.67%)	16/7 (69.57%)	0.05[c]	1.000
Hachinski ischemia scale	1.38 (0.17)	1.13 (0.10)	−1.01[a]	0.313
Hypertension disease (%)	2/22 (8.33%)	1/22 (4.35%)	0.31[c]	1.000
Diabetes mellitus (%)	1/23 (4.17%)	1/22 (4.35%)	0.00[c]	1.000
Chinese Smell Identification Test	19.58 (6.83)	18.3 (5.55)	−0.74[a]	0.461
Baseline scores on Associative Memory—Primary outcomes				
Recall	1.56 (1.42)	1.39 (1.12)	−0.04[a]	0.971
Baseline scores on clinical outcome measures—Secondary outcomes				
Montreal Cognitive Assessment	13.96 (5.61)	14.65 (5.45)	−0.43[b]	0.669
Mini-Mental State Examination	20.54 (4.67)	21.7 (4.67)	−0.85[b]	0.401
Hamilton Anxiety Rating Scale	5.5 (4.82)	5.39 (6.79)	−0.77[a]	0.441
Hamilton Depression Rating Scale	4.71 (3.9)	3.87 (4.86)	−1.25[a]	0.213
Neuropsychiatric Inventory	8.54 (5.93)	8.96 (6)	−0.52[a]	0.600
ADL	26.71 (5.78)	28.52 (8.75)	−0.68[a]	0.498
Clinical Dementia Rating	1.96 (2.18)	2.65 (3.35)	−0.41[a]	0.685
Baseline scores on Multi-domain Cognition Assessment—Secondary outcomes				
Memory function assessment				
Logical Memory Test-Immediate	7.63 (6.77)	7.26 (6.82)	0.18[b]	0.855
Logical Memory Test-Delay	4.63 (6.20)	2.7 (6.53)	−1.46[a]	0.144
AVLT-Immediate	3.29 (2.61)	3.43 (1.69)	−0.22[b]	0.824
AVLT-Delay	2.50 (3.45)	1.04 (1.69)	−1.58[a]	0.113
AVLT-Recognition	12.13 (2.35)	11.70 (3.32)	0.51[b]	0.610
Attention function assessment				
Digital span test-Forward	5.71 (1.71)	5.70 (1.11)	−0.38[a]	0.705
Digital span test-Backward	3.29 (1.40)	3.13 (0.92)	−0.86[a]	0.389
Symbol Digit Modalities Test	11.17 (7.99)	14.65 (12.19)	−1.15[b]	0.256
Executive function assessment				
Stroop-Dot (s)	46.90 (28.86)	38.18 (36.02)	−1.45[a]	0.148
Stroop-Word (s)	62.73 (35.35)	70.35 (68.63)	−0.75[a]	0.456
Stroop-Color Word (s)	89.6 (51.73)	101.21 (97.43)	−0.83[a]	0.406
Color Trail Test-Reaction Time-A (s)	224.95 (167.42)	189.02 (104.39)	−0.40[a]	0.687
Color Trail Test-Reaction Time -B (s)	1.14 (2.26)	1.52 (3.50)	−0.83[a]	0.406
Visuospatial function assessment				
Clock Drawing Test	1.5 (0.83)	1.87 (1.01)	−1.48[a]	0.138
Hooper Visual Organization Test	10.31 (5.74)	10.61 (4.58)	−0.19[b]	0.846
Judgment of Line Orientation Test	18.46 (6.40)	19.91 (6.73)	−0.88[a]	0.376
Language function assessment				
Boston Naming Test-Total	17.08 (5.85)	17.09 (4.47)	−0.02[b]	0.998
Boston Naming Test-Spontaneous	16.46 (5.63)	16.70 (4.33)	−0.16[b]	0.873
Verbal Fluency Test-Sematic	9.71 (4.36)	10.70 (4.27)	−0.78[b]	0.437
Verbal Fluency Test-Letter	4.17 (3.51)	4.26 (3.33)	−0.09[b]	0.925

Abbreviations: ADL = Activities of Daily Living scale; AVLT = Chinese version of the Auditory Verbal Learning Test; NPI = Neuropsychiatric Inventory.
Note: * Values in brackets indicate standard deviation.
[a] Mann-Whitney *U* test.
[b] Two-sample *t*-test.
[c] Chi-squared test.

displayed higher AM scores than the sham group at weeks 2 (mean difference = 1.35, $t = 2.60$, $P = 0.014$, effect size = 0.75; 95% CI, 0.15−1.36) and 10 (mean difference = 1.07, t = 2.55, $P = 0.016$, effect size = 0.69; 95% CI, 0.07−1.30).

3.4. Secondary outcomes

The responder ratio was higher among the active group ($n = 11$, 45.83%) than the sham group ($n = 2$, 8.69%) at week 2 ($\chi^2 = 8.09$, $P = 0.004$; Fig. 2B). There was also a marked difference at week 10 (33.33% in active versus 4.76% in sham, $\chi^2 = 5.71$, $P = 0.025$; Fig. 2B).

For symptom measures, group (active and sham) by time (baseline and weeks 2 and 10) interaction effects (Fig. 2C–D, Table 2) were observed in the MoCA ($P < 0.001$), MMSE ($P < 0.001$), ADL scores ($P = 0.040$), HAMA ($P < 0.001$), and HAMD ($P < 0.001$), but not the NPI ($P = 0.19$). Furthermore, all symptoms were lower in the active group than in the sham group after treatment ($P < 0.05$; Table e1-2).

For multi-domain cognition tests, significant group (active and sham) by time (baseline and weeks 2 and 10) interaction effects were observed in most of the memory (LMT-Immediate/Delay, AVLT-Immediate/Delay/Recognition), attention (DST-F/B, SDMT), language (BNT, VFT-S/L), executive function (SCWT-Dot), and visual-spatial function (CDT, HVOT, and JOLT) assessments ($P < 0.05$; Fig. 2C–D, Table 2). Most tests revealed improved scores in the active group compared with the sham group after treatment (Table e1-2). Three tests (SCWT-word, SCWT-color word, and CTT) did not show significant interaction effects.

3.5. Correlations between clinical variables

Pearson's correlation revealed that AM scores were significantly correlated with MoCA, MMSE, ADL, and CSIT scores (all $P < 0.05$, Table e3) at baseline. Furthermore, the higher MMSE score at baseline was associated with greater AM improvement at weeks 2 ($r = 0.716$, $P < 0.001$) and 10 ($r = 0.755$, $P < 0.001$).

Table 2
Estimated mean score and effect sizes in primary and secondary outcome variables.

	Time × Group[a]			The score at week 2 [b]		The score at week 10 [b]	
	$F_{(2,87)}$	*P*	*Partial* η^2	Active (n = 24)	Sham (n = 23)	Active (n = 23)	Sham (n = 21)
Associative Memory	12.65	<0.001	0.225	2.54 (2.30)	1.20 (1.03)	2.24 (2.02)	1.17 (0.94)
MoCA	19.07	<0.001	0.305	16.75 (6.72)	14.39 (5.84)	15.48 (7.13)	13.29 (5.60)
MMSE	16.40	<0.001	0.274	22.83 (5.39)	20.96 (5.17)	21.52 (6.33)	20.24 (5.04)
HAMA	8.46	<0.001	0.163	2.96 (2.40)	5.87 (7.15)	3.22 (2.58)	5.19 (5.97)
HAMD	13.36	<0.001	0.235	1.58 (2.04)	4.87 (4.89)	3.00 (2.71)	3.95 (4.20)
NPI	1.69	0.190	0.037	2.21 (2.67)	7.00 (8.91)	5.61 (8.89)	7.62 (10.74)
ADL	3.34	0.040	0.071	25.17 (5.36)	28.65 (8.38)	26.65 (6.97)	29.48 (8.89)
LMT-Immediate	22.14	<0.001	0.337	11.88 (8.31)	5.00 (4.97)	10.13 (7.61)	4.62 (4.01)
LMT-Delay	18.35	<0.001	0.297	8.54 (7.66)	2.43 (3.95)	7.74 (7.04)	1.76 (3.48)
AVLT-Immediate	9.86	<0.001	0.185	5.10 (3.13)	3.57 (1.87)	4.50 (2.91)	3.30 (1.58)
AVLT-Delay	11.60	<0.001	0.211	5.46 (4.48)	1.35 (1.07)	4.30 (3.70)	1.48 (1.08)
AVLT-Recognition	9.21	<0.001	0.175	13.63 (2.08)	11.35 (3.42)	13.52 (2.13)	11.19 (3.11)
DST-Forward	8.83	<0.001	0.169	6.50 (1.67)	5.70 (0.97)	6.43 (1.95)	5.29 (1.01)
DST-Backward	4.82	0.010	0.100	3.88 (1.23)	3.00 (0.85)	3.61 (1.56)	3.05 (0.92)
SDMT	12.63	<0.001	0.225	14.54 (10.03)	13.61 (11.60)	12.74 (9.87)	13.57 (11.11)
SCWT-Dot (s)	7.41	<0.001	0.145	37.03 (22.49)	42.03 (37.96)	41.85 (28.29)	44.45 (38.46)
SCWT-Word (s)	1.89	0.158	0.042	53.28 (30.13)	72.22 (72.89)	58.75 (42.28)	75.44 (68.94)
SCWT-Color Word (s)	2.22	0.115	0.049	76.24 (50.00)	101.83 (100.76)	79.92 (52.45)	99.19 (92.79)
CTT-A (s)	0.26	0.771	0.006	200.05 (136.26)	175.46 (107.77)	226.88 (156.77)	212.20 (122.45)
CTT-B (s)	0.47	0.624	0.012	398.54 (264.78)	401.43 (310.64)	325.99 (279.64)	363.15 (257.60)
CDT	6.26	0.003	0.126	2.54 (0.93)	2.00 (0.74)	2.13 (0.97)	1.90 (0.77)
HVOT	5.02	<0.001	0.103	13.40 (6.86)	10.30 (5.30)	11.24 (5.97)	9.76 (4.99)
JOLT	13.73	0.009	0.24	21.85 (6.07)	19.50 (7.15)	19.37 (7.67)	18.19 (8.03)
BNT-Total	22.24	<0.001	0.340	20.83 (5.53)	16.39 (4.99)	20.30 (6.26)	16.29 (4.96)
BNT-Spontaneous	9.24	<0.001	0.175	19.71 (5.81)	16.26 (4.93)	18.35 (6.42)	16.14 (4.96)
VFT-Semantic	4.58	0.013	0.095	12.00 (5.30)	10.70 (4.54)	11.22 (5.65)	9.62 (4.15)
VFT-Letter	5.85	0.004	0.119	5.13 (2.85)	3.83 (2.57)	4.70 (3.08)	4.48 (3.06)

Note: Values in brackets indicate standard deviation. P-values are false discovery rate corrected. [a] interaction effect in linear mixed effect models indicating outcome changes (from baseline to weeks 2 and 10) between groups (Active and Sham); [b] the positive value represents improvement and a negative value represents deterioration.
Abbreviations: MoCA: Montreal Cognitive Assessment; MMSE: Mini-Mental State Examination; HAMA: Hamilton Anxiety Rating Scale-14 items; HAMD: Hamilton Depression Rating Scale-17 items; NPI: Neuropsychiatric Inventory; ADL: Lawton-Brody Activities of Daily Living scale; CDR: Clinical Dementia Rating; LMT: Logical Memory Test; CAVLT: Chinese version of the Auditory Verbal Learning Test; DST: Digital Span Test; SDMT: Symbol digit modalities test; SCWT: Stroop Color-Word Test, CTT: Color Trail Test; CDT: Clock-Drawing Test; HVOT: Hooper Visual Organization Test; JLOT: Judgment of Line Orientation Test; BNT: Boston Naming Test; VFT: Verbal Fluency Test.

There was no marked association between the AM changes and HAMA ($r = -0.264$, $P = 0.212$) or HAMD changes ($r = -0.226$, $P = 0.289$) at week 2 among the active group, indicating that changes in anxiety or depression did not cause improvements in AM.

Spearman correlation analysis demonstrated that there was a positive correlation between the change of AM and BNT-Total (r' = 0.437, P = 0.037) at week 2, but not week 10 ($P > 0.05$; Fig. S2).

3.6. Baseline MMSE predicted AM improvements

Multivariate linear regression ($R^2 = 0.537$, $F = 25.57$, $P < 0.001$) indicated that the baseline MMSE ($P < 0.001$) but not MoCA, ADL, or CSIT scores ($P > 0.05$), significantly predicted AM changes in the active group at week 2 (Table e4).

Logistic regression indicated that the MMSE ($\beta = 0.88$, $P = 0.016$) was the most important variable in predicting response to treatment, excluding age, sex, education, HIS, CSIT, baseline MoCA, baseline MoCA, NPI, ADL, CSIT, HAMA, or HAMD (Table e5).

The regression model between baseline MMSE and AM changes was used to predict AM changes and responders in an independent cohort ($n = 13$). Pearson's correlation analysis indicated a significant correlation between the predicted and actual AM changes ($R^2 = 0.88$, $P < 0.001$; Fig. 3A). The logistic regression was able to identify the responders with a 92.31% accuracy (sensitivity = 86%, specificity = 100%; Table e6, Fig. 3B).

3.7. Subgroup analysis determined by the MMSE

Generally, an MMSE score of 21–24 is defined as mild dementia [53]. Considering the predictive ability of MMSE, all patients were separated into high (High MMSE [HM] ≥ 21, $n = 30$; Active-HM 18, Sham-HM 12) and low MMSE groups (Low MMSE [LM] < 21, $n = 30$; Active-LM 19, Sham-LM 11), and analyzed using linear mixed-effect models nested within individuals. Patients with HM were similarly distributed between active and sham groups (45.83% versus 52.17%, $\chi^2 = 0.19$, $P = 0.664$).

There was a significant interaction effect between time (baseline and weeks 2 and 10) and groups on AM scores ($F_{(4,98)} = 7.85$, $P < 0.001$; Table e7). The AM change at weeks 2 and 10 were significant in the active-HM group, but not in the active-LM group. The AM scores were highest in the active-HM group at weeks 2 and 10 (Table e8-9, Fig. 4A). The responder–non-responder ratio in the active-HM group was higher than the active-LM (week 2, $\chi^2 = 11.91$, $P = 0.001$; week 10, $\chi^2 = 6.41$, $P = 0.014$) and sham groups ($P < 0.001$; Fig. 4B). Similarly, the improvement in the active-HM group was greater than the active-LM group in most of the symptom measures (i.e., MoCA, MMSE, ADL, HAMA, and HAMD; all $P < 0.05$; Fig. 4C–D) and multi-domain cognition tests (i.e., LMT-Immediate/Delay, AVLT-Immediate/Delay/Recognition; SDMT; CDT, HVOT, and JOLT; BNT, VFT-S/L; all $P < 0.05$; Fig. 4C–D). More details can be found in Table e8-9.

3.8. Adverse effects

Eight participants reported adverse events: five patients reported painful scalp sensations (three in active and two in sham), two patients in the active group reported eyelid twitches, and one patient in the sham group reported tinnitus. All the events were tolerable, and all diminished gradually after treatment cessation. No epilepsy or epileptic seizures were recorded.

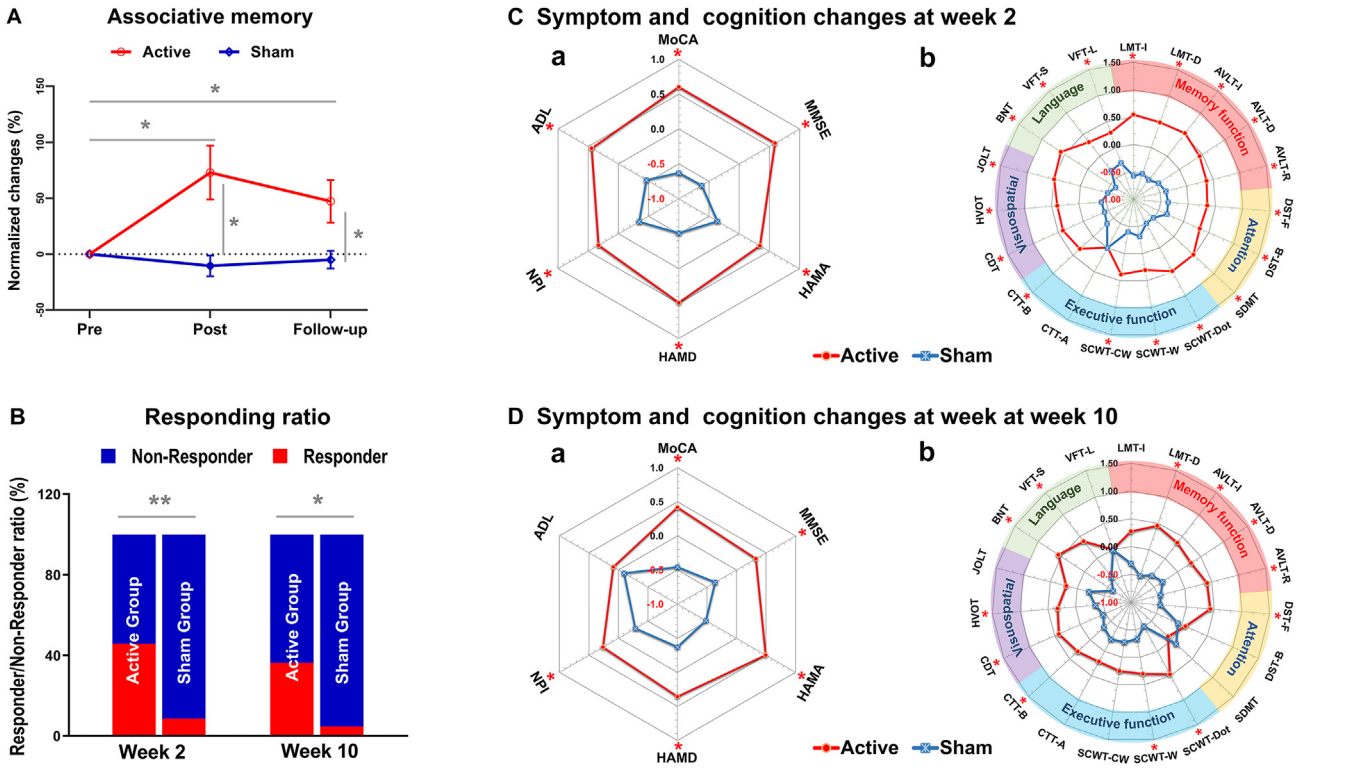


Fig. 2. Aftereffects of rTMS treatment in the active and sham groups. The active group showed greater AM improvements (A; mean ± SE) and responder-non-responder ratio (B) than the sham group at weeks 2 and 10. All symptom and cognitive tests (except CTT-A) demonstrated greater improvements in the active than the sham group at week 2 (C). At week 10, 17 of the 26 symptoms or cognitive tests showed a larger improvement in the active than the sham group (D). Red stars indicate tests showing different changes at weeks 2 (C) and 10 (D). All data at weeks 2 and 10 were normalized changes relative to the baseline scores. The vertical axis of the radar map is the percentage of symptom improvement after z conversion. Abbreviations: ADL = Activities of Daily Living scale; AM = Associative Memory; AVLT = Chinese version of the Auditory Verbal Learning Test; BNT = Boston Naming Test total; CDT = Clock-Drawing Test; CTT = Color Trail Test; DST = Digital Span Test; HAMD = Hamilton Depression Rating Scale-17 items; HVOT = Hooper Visual Organization Test; JLOT = Judgment of Line Orientation Test; LMT = Logical Memory Test; MoCA = Montreal Cognitive Assessment; MMSE = Mini-Mental State Examination; NPI = Neuropsychiatric Inventory; SCWT = Stroop Color-Word Test, SDMT = Symbol digit modalities test; VFT = Verbal Fluency Test. *$P < 0.05$; **$P < 0.01$. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

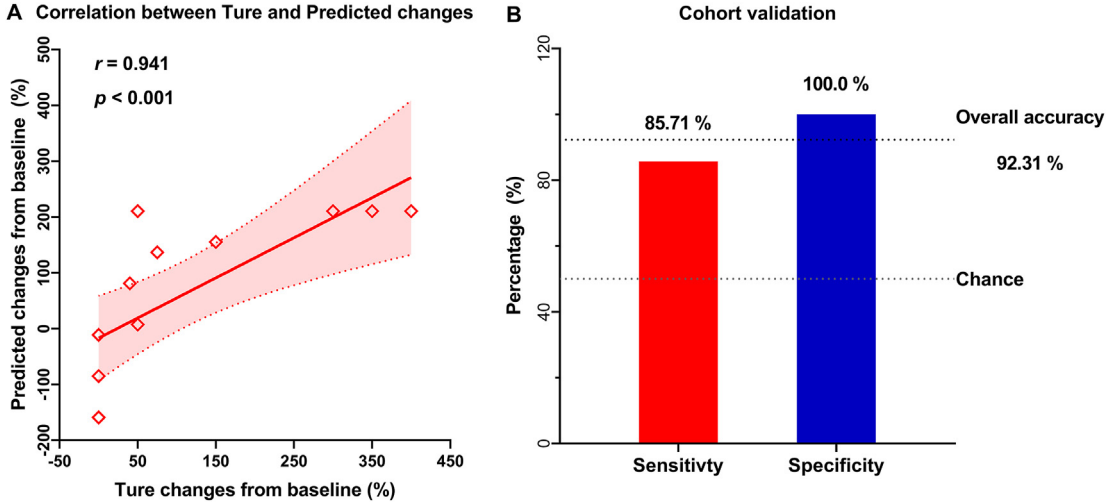


Fig. 3. Predicting the AM of an independent cohort after active rTMS treatment. The actual AM changes in AD patients are highly correlated to the predicted changes (A), computed using the linear regression model based on this RCT data. The logistic regression model can identify responders in the independent cohort with a 92.31% accuracy (B).

4. Discussion

Using accelerated iTBS, the present RCT indicated that the AM deficits of patients with AD could be markedly ameliorated after 14 days of treatment. Additionally, there were three secondary findings: 1) the improvement of AM can persist for 2 months after the end of treatment; 2) baseline MMSE positively predicted the AM improvement, which was validated by independent data; and 3) both clinical symptoms (e.g., MoCA and MMSE) and multi-domain cognitions markedly improved after active treatment, which was more prominent among patients with a high baseline MMSE. These findings suggest that accelerated iTBS can effectively improve AM

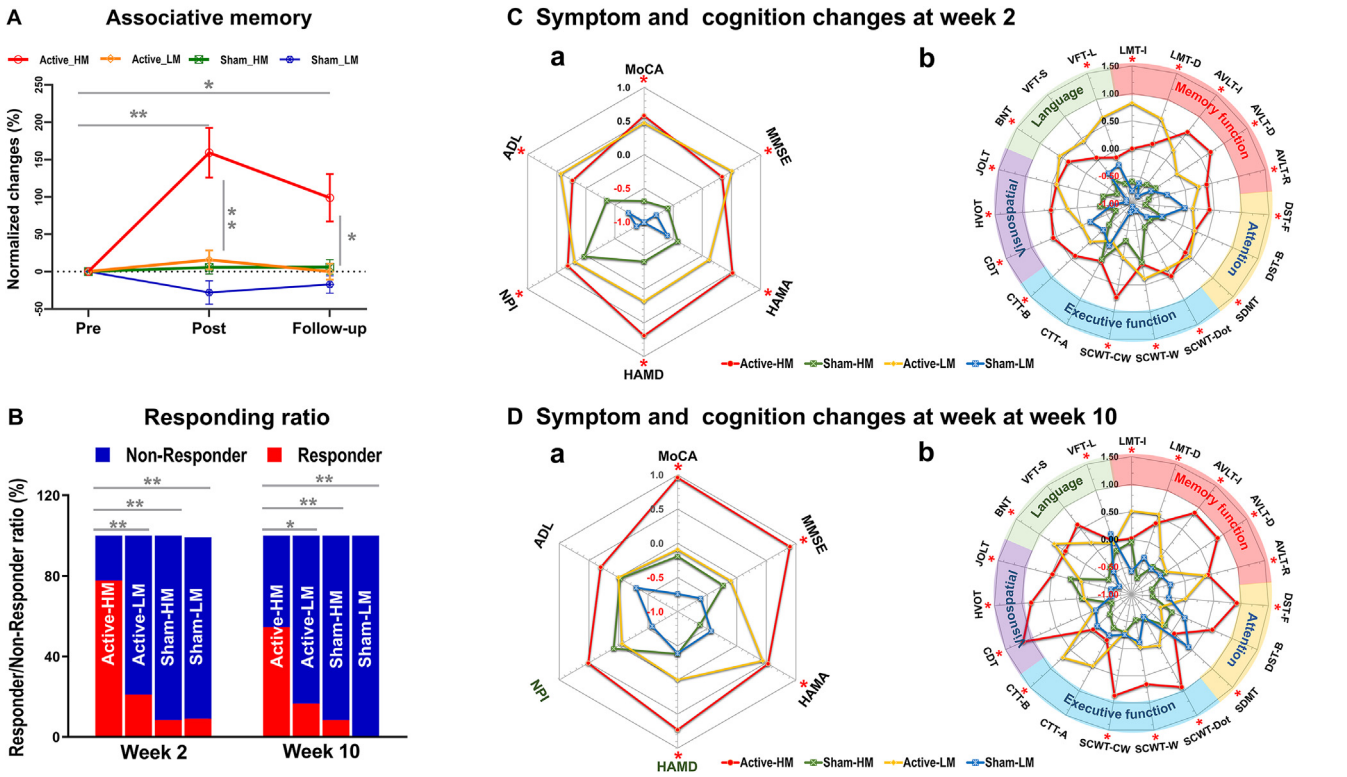


Fig. 4. Aftereffects of rTMS treatment in the Active-HM, Active-LM, Sham-HM, and Sham-LM groups. The Active-HM group showed greater AM improvements (A; mean ± SE) and responder-non-responder ratio (B) than other groups at weeks 2 and 10. The AM scores of the Active-HM group were also higher than other groups at weeks 2 and 10 (A). Red stars indicate tests showing different changes at weeks 2 (C) and 10 (D). All data at weeks 2 and 10 were normalized changes relative to the baseline scores. The vertical axis of the radar map is the percentage of symptom improvement after z conversion. Abbreviations: ADL = Activities of Daily Living scale; AM = Associative Memory; AVLT = Chinese version of the Auditory Verbal Learning Test; BNT = Boston Naming Test total; CDT = Clock-Drawing Test; CTT = Color Trail Test; DST = Digital Span Test; HAMD = Hamilton Depression Rating Scale-17 items; HVOT = Hooper Visual Organization Test; JLOT = Judgment of Line Orientation Test; LMT = Logical Memory Test; MoCA = Montreal Cognitive Assessment; MMSE = Mini-Mental State Examination; NPI = Neuropsychiatric Inventory; SCWT = Stroop Color-Word Test, SDMT = Symbol digit modalities test; VFT = Verbal Fluency Test. $^{*}P < 0.05$; $^{**}P < 0.01$. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

as well as the symptoms and other cognitive function among patients with AD, especially for individuals with relatively high MMSE scores.

AM is a vital function in everyday life. Many rTMS studies have aimed to enhance this function in healthy individuals and patients with AD. For healthy individuals, a series of studies have shown the potential for high frequency rTMS to improve AM [11,54,55]. More recently, rTMS in patients with AD was observed to significantly improve AM at 8 weeks following a 4-week treatment schedule with cognitive training [56]. A similar improvement was reported in our previous open-label study, which utilized only 2 weeks of accelerated iTBS without cognitive training [43]. Here, the same protocol was adopted and further demonstrated that AM improvements were greater than the placebo effect and could persist for 8 weeks. Thus, compared with previous similar studies [56,57], our rTMS protocol significantly accelerated the treatment course and simplified the procedure while maintaining treatment efficiency. Of the 24 patients in the active group, 11 (46%) were responders. For eight of them, the benefit persisted for more than 8 weeks. This high efficiency may be partly ascribed to two factors. First, the iTBS mimicked the endogenous theta-gamma rhythms in the memory system [30], which may be inherently powerful for enhancing memory function. Second, high-resolution structural images were used to precisely stimulate the target on each treatment day. Compared with the 10–20 electrode system, this neuronavigational approach can more easily induce the same behavior effect [41].

To further improve clinical efficiency, one approach is to identify individualized outcome predictors. Among baseline measures, we observed that the MMSE was positively correlated with AM improvements. Consistent with previous AD studies [56–58], our findings indicate that less impaired patients are more likely to benefit from rTMS treatment. We further demonstrate the reliability of the predictors in independent samples and the responders could be identified with a 92% accuracy. In the subgroup analysis, the HM group also displayed greater AM improvements than the LM group; however, the latter did not show greater changes than the sham group. This outcome difference may be because the neuroplasticity of the DLPFC is better among patients with less impaired frontal function; thus, it was more easily modulated by rTMS treatment. For future clinical application, this iTBS protocol is recommended for patients with mild or moderate stages of AD [59], especially those with MMSE scores ≥21.

Furthermore, the improvements we observed in other neuropsychology assessments of AD, including the global cognition, ADL, moods, memory, attention function, and language function, were consistent with previous studies [59–61]. Particularly, MMSE improvements were greater than the clinically significant minimum difference (1.4 points) [62,63] and was maintained for longer than 8 weeks. Notably, the effects in most cognition tests were consistent with previous studies [26,64–66], but more prominent in the active-HM group. The psychological symptoms (i.e., HAMA, HAMD) were significantly alleviated and persisted until week 10. These changes likely occurred because the DLPFC is a hub region of the

mood regulation network [67]. The results suggest that our rTMS protocol may also be effective for senile depression [68]; however, changes of emotion unrelated to memory improvement indicate that there may be a mechanism separating the two, and DLPFC has an important regulatory role between emotion and cognition.

The improvements of AM and clinical symptoms after iTBS may be due to the broad functions of the DLPFC [69,70]. Former studies have shown that the DLPFC is a key part of the executive control and frontoparietal networks, which have important roles in multiple cognitive functions [10,69–72]. Many studies have also demonstrated that iTBS in the DLPFC may heighten cortical excitability, thereby enhancing synaptic plasticity [28,29]. By promoting synaptic plasticity, efficiency of the DLPFC is increased, which has subsequent effects on improving cognitive function, such as AM, language function, semantic memory, and working memory [73]. A previous study demonstrated that the hyperactivity of the left DLPFC was beneficial to the encoding performance in language and memory processing [21,34,74–77]. Therefore, improvements in cortical excitability and encoding function of the DLPFC caused by iTBS may be a potential neuro-mechanism of memory enhancement in AD, which has been corroborated by other rTMS-related studies [26,74,78].

Our present study demonstrated that improved AM is related to changes in naming ability. This may be due to both the hippocampus and DLPFC having critical roles in memory and language processing, which may serve as a crossroad between language and memory [76,79–81]. Prior studies suggest that left DLPFC activation and its improved connectivity with the hippocampus may reflect processes that are important for enhancing memory [21,75] and language [34,76]. Hence, the increased excitability of the left DLPFC caused by iTBS may be beneficial to memory and language functions; however, in the present AM task, language and memory may be difficult to separate. In the future, additional research is needed to further explore possible mechanisms.

The present study has several limitations. First, although the sample size was defined a priori, it was relatively small, suggesting that our findings should be confirmed with a large-scale, multicentered clinical trial. Second, the neuronavigational system was used to precisely identify the target based on individualized anatomical features; however, the DLPFC was one of the cortices with high functional variability across participants [82]. To further improve the treatment efficiency, future studies should define the target using functional features [83,84] and estimate the induced electronic field individually [85]. Third, likely due to the relatively short course of treatment, the aftereffects on some cognitive functions decreased markedly at follow-up. It would be interesting to test whether repeating the iTBS protocol every 2 months could prolong the treatment effect for years. Fourth, the diagnosis of AD in our study was based on clinical assessment and did not incorporate pathologic markers of AD (e.g., amyloid or tau imaging). Finally, the total duration of AM was relatively short (<5 min). As a result, the AM score may largely reflect the encoding function of participants. estimated in our current experimental design, the assessments used are more a test of encoding than recall [86]. Actually, in any given cognitive task, encoding and retrieval may be difficult to separate completely, an online-TMS task design will be needed in the future to better explain the effects on AM task in detail.

5. Conclusions

In this RCT, we observed that accelerated iTBS of the left DLPFC effectively alleviated deficits in the encoding function of AM, which was sustained for 8 weeks after treatment cessation. Notably, the baseline MMSE scores were able to predict treatment outcomes. The ameliorating effects were more effective and robust among patients with mild AD (MMSE ≥21) than those with severe AD. In summary, our findings provide RCT evidence for the potential use of iTBS in the encoding function of AM and cognitive enhancement among patients with AD. Future studies should investigate the efficacy across multiple centers and characterize the neural mechanisms involved in these improvements.

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Data availability

The data that support the findings of this study are available from the corresponding authors upon request.

CRediT authorship contribution statement

Xingqi Wu: Conceptualization, Methodology, Formal analysis, Data curation, Writing – original draft, Visualization. **Gong-Jun Ji:** Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing – review & editing. **Zhi Geng:** Investigation. **Lu Wang:** Investigation, Visualization. **Yibing Yan:** Investigation. **Yue Wu:** Investigation. **Guixian Xiao:** Investigation. **Liying Gao:** Investigation. **Qiang Wei:** Investigation, Funding acquisition. **Shanshan Zhou:** Investigation, Supervision, Data curation. **Ling Wei:** Investigation, Supervision, Funding acquisition. **Yanghua Tian:** Conceptualization, Supervision, Project administration, Writing – review & editing. **Kai Wang:** Conceptualization, Supervision, Project administration, Funding acquisition, Writing – review & editing.

Declarations of interest

None.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.brs.2021.11.007.

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